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EXTENDED ABSTRACT

Learning Spatiotemporal Precipitation Fields with Probabilistic Neural Processes

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1. Introduction and Objectives

Reliable, high resolution precipitation fields are central to hydrology, flood risk assessment, agriculture, and climate impact studies, yet remain difficult to construct from sparse and irregular rain gauge networks in the United Kingdom (UK) and elsewhere [1]. While reanalysis products such as ERA5 provide physically consistent gridded estimates through data assimilation, their effective resolution, temporal granularity, and reliance on sparse observations limit their suitability for local and near real time applications. Recent advances in probabilistic machine learning (ML), particularly neural processes, offer a principled alternative for spatiotemporal interpolation under irregular sampling by combining uncertainty quantification with scalable representation learning [2, 3].

Objectives. By casting precipitation interpolation as learning a conditional distribution over rainfall at different spatiotemporal sites, conditioned on rain gauge observations split into context and target sets, we assess whether neural process based models can outperform reanalysis products such as ERA5 for rainfall interpolation at local to regional scales. Using high quality UK rain gauge observations, we develop interpolated precipitation fields using convolutional neural processes implemented in DeepSensor [4, 5]. Performance is evaluated against reanalysis and conventional methods such as Kriging, with a focus on uncertainty calibration, robustness to sparse observations, and the ability to generate sub-hourly estimates. Additionally, we examine few shot and zero shot transfer across regions with differing climatological regimes, addressing generalisation and data efficiency. Overall, this work aligns with the aims of climate informatics by developing data driven approaches that complement physical models and support rigorous, reproducible evaluation across ML and climate science [6].

2. Methods and Data

We implement and compare two Neural Process (NP) variants: an existing DeepSensor Convolutional Conditional Neural Process (ConvNP) and a novel Transformer-based NP [7]. The models map a context set, comprising full auxiliary grids and a random subset of precipitation sensors—to a predictive distribution at target locations as described in Fig. 1. To address the zero-inflated, heavy-tailed nature of

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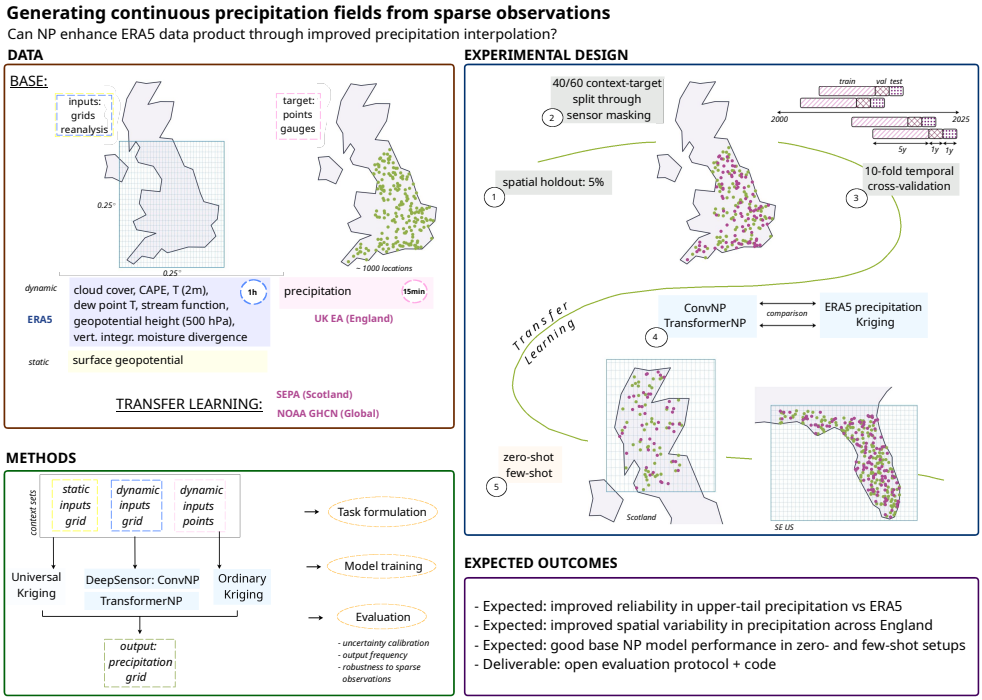

Figure 1. *General overview of the spatiotemporal precipitation interpolation framework.*

rainfall, we employ a Bernoulli-Gamma likelihood. For training, we utilise a random sensor-masking strategy to enable zero-shot generalisation to unseen coordinates, and a 10-fold cross-validation scheme, additionally evaluated via transfer learning experiments across climatological regimes. Heterogeneous data sources: dense gridded auxiliary variables from ERA5 reanalysis (0.25°, hourly) and sparse point-based precipitation observations. Input auxiliary variables include static surface geopotential and dynamic atmospheric descriptors (e.g., specific humidity, temperature, convective available potential energy (CAPE), convective inhibition (CIN), etc.).

3. Expected Outcomes

This study delivers a quantitative assessment of NP models for spatiotemporal interpolation of precipitation. The primary outcome aims for a comparison between ConvNP and TransformerNP interpolation, ERA5 reanalysis precipitation, and conventional interpolation methods, conducted using identical training and evaluation periods and assessed over RMSE, and uncertainty quantification of precipitation values (both usual scenarios and extremes). This study further evaluates model generalisation through few-shot transfer experiments across regions with differing climatological and observational characteristics, including England, Scotland, and selected regions from NOAA GHCN in the US, supporting the role of NP-based methods as complementary, data-driven alternatives to reanalysis outputs.

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